

API Reference

KMS to SDN Agent API

API Version: 2.0.0

Messages the KMS sends to the SDN

CONTACT

URL: <https://github.com/ait-crypto/kms-sdn-agent-interface-specification/issues>

INDEX

1. API	4
1.1 GET /version	4
2. LINK	5
2.1 POST /link/registration	5
2.2 PATCH /link/registration	6
2.3 DELETE /link/{key_stream_id}	7
3. NOTIFICATION	8
3.1 POST /notification/registration	8
3.2 PATCH /notification/status	9
3.3 PUT /notification/event	9

Security and Authentication

SECURITY SCHEMES

KEY	TYPE	DESCRIPTION
https	mutualTLS	Connections must use TLS 1.3. The server enforces this at the transport layer.

API

1. API

1.1 GET /version

Get this API version

State this SDN API version number which is implemented and supported by the KMS

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Provided API version

RESPONSE MODEL - application/json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
version*	string	PATTERN: ^(0 [1-9]\d*)\.(0 [1-9]\d*)\.(0 [1-9]\d*)(?:-((?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)*)?)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)*)?)?\$ Version number in semantic versioning

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

2. LINK

2.1 POST /link/registration

Post new link registration

When a source APP registers with a KMS, the KMS sends the parameters supplied by the APP to the SDN. The SDN configures a path between the source and destination, verifies the QoS and returns the key stream id and adjacent node information

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
source*	string	URI or UUID of the source APP
destination*	array	
qos_request*	object	
type*	enum	ALLOWED: etsi_004, etsi_014
etsi_014	object	
number*	integer	Number of keys
size*	integer	Size of keys in bit
extensions	array	
etsi_004	object	
key_chunk_size*	integer	length of key chunks in Byte.
max_bps*	integer	
min_bps*	integer	
jitter	integer	
priority	integer	
timeout	integer	
ttl*	integer	
metadata_mimetype	string	
performance*	array	
link_id	string	
current_kpi	{recursive}	kms-performance-entry
options	object	
multipath	boolean	In case of a multi-path establishment, specify 'true'.

RESPONSE

STATUS CODE - 201: Successful registration

RESPONSE MODEL - application/json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
key_stream_id*	string	
next_node_info*	array	

NAME	TYPE	DESCRIPTION
qos	object	

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

2.2 PATCH /link/registration

Update link status by destination APP

When a destination APP registers with a KMS, the KMS informs the SDN of the destination APP'S registration. This is required in ETSI 015.

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
key_stream_id*	string	
destination*	string	Of type UUID or URI

RESPONSE

STATUS CODE - 200: Successful registration

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found.

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

2.3 DELETE /link/{key_stream_id}

Delete key stream

If the ETSI 004 application closes a key stream, this is reported to the SDN

REQUEST

PATH PARAMETERS

NAME	TYPE	EXAMPLE	DESCRIPTION
*key_stream_id	uuid		

RESPONSE

STATUS CODE - 200: Link closed

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found.

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

3. NOTIFICATION

3.1 POST /notification/registration

Post KMS added to network

Inform the SDN of a new KMS that has been added to the network. If local instance has information about connected QKD devices or links, specify in request body.

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
kms_id*	string	
model*	object	
device_vendor*	string	Vendor of the KMS
device_model*	string	Device brand name
device_version*	string	version number in semantic versioning
qkd_device_ids	array	
link_ids	array	

RESPONSE

STATUS CODE - 201: Registration received. Optionally additional config data provided in response. Config file data format or content not defined and expected to be vendor specific.

RESPONSE MODEL - application/octet-stream

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

3.2 PATCH /notification/status

Update KMS status

Inform the SDN of when the KMS's status changes, e.g. the KMS goes into maintenance mode

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
state*	string	
reason*	string	

RESPONSE

STATUS CODE - 200: Status received

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

3.3 PUT /notification/event

Update event at KMS

Inform the SDN of an event

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
level*	enum	ALLOWED:debug, info, warning, error, critical
code	integer	
reason*	string	

RESPONSE

STATUS CODE - 200: Event information received

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
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